

Collective Moral Disengagement as Evolutionarily Stable Strategy (ESS): Extended Phenotype Theory and the Intra-Corporate Intentionality Paradox

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Abstract

The literature on collective moral disengagement (CollMD) explains how organizations construct shared narratives that justify unethical behavior, but it does not specify under what conditions that state is evolutionarily stable, nor why it persists beyond the individuals who generated it. This article applies Extended Phenotype Theory (EPT) and multilevel Evolutionary Game Theory (EGT) to reinterpret CollMD as an evolutionarily stable strategy (ESS) in organizational environments with high internal normative rigidity, measured through an Organizational Lock-in Index (OLI) derived analogously from the Constitutional Lock-in Index (Lerer, 2025a). The article argues that CollMD is not merely emergent but actively selected because it builds extended phenotypes: performance evaluation systems, reporting structures, and accountability-diffusing narratives that reproduce the collective moral state independently of individual turnover. The transmission mechanism is formalized through Heteronomous Bayesian Updating (HBU), which predicts that the speed of CollMD adoption correlates with the visibility of reactions to dissent rather than with the argumentative quality of available moral justifications. CollMD is further identified as the psychosocial mechanism that resolves the intra-corporate intentionality paradox of Asymmetric Intentionality Theory (AIT): it enables Level 3 individuals to operate within Level 1 organizations by collectively reconstructing the moral space until both levels appear compatible. The framework is applied qualitatively to four documented cases (Volkswagen, Siemens, Wells Fargo, and Purdue Pharma), identifying OLI trajectories, extended phenotype mechanisms, and tipping points consistent with EGT predictions. Analysis suggests that OLI exceeds 0.65 in all four cases before unethical behavior becomes systematic, that CollMD extended phenotypes are detectable between 18 and 36 months before visible risk materializes, and that effective intervention requires altering the architecture of visible reactions rather than the content of circulating narratives.

Keywords: collective moral disengagement, evolutionarily stable strategy, extended phenotype theory, organizational lock-in, heteronomous Bayesian updating, asymmetric intentionality, compliance, corporate governance.

I. INTRODUCTION

Organizations populated by morally ordinary people produce collectively immoral behavior with sufficient regularity to constitute a structural phenomenon rather than an exceptional deviation. Volkswagen employed approximately 600,000 people when engineers and managers

coordinated the concealment of nitrogen oxide emissions in approximately 11 million vehicles over nine years. Purdue Pharma maintained a board of directors, a legal department, and a compliance function while systematically misleading physicians and regulators about the addictive properties of OxyContin. Wells Fargo maintained a code of ethics and an internal whistleblower hotline while approximately 5,300 employees opened roughly 3.5 million unauthorized customer accounts. These cases share a structural feature that neither individual bad-actor explanations nor organizational pressure narratives can fully account for: the transition from isolated deviance to institutionalized wrongdoing.

The literature on moral disengagement has progressively elaborated the mechanisms through which this transition occurs. Bandura (1999, 2016) identified the psychological mechanisms by which individuals selectively disengage moral standards: moral justification, displacement of responsibility, diffusion of responsibility, and minimization of consequences, among others. More recently, scholarship has shifted toward collective analysis. Schaefer et al. (2026) introduce a multilevel conceptualization of collective moral disengagement (CollMD) as an emergent state produced through a three-phase process of individual sensemaking, sensegiving, and collective sensemaking, operating through recursive feedback between individual and group levels. Their contribution establishes CollMD as a group-level phenomenon with complex systems dynamics, not a simple aggregation of individual moral failures.

The present article argues that this emergentist framing, while necessary, is insufficient. Emergence explains origin; evolutionary stability explains persistence. CollMD persists across personnel turnover, resists ethics training, and reproduces itself in new organizational members who had no role in its construction because it builds extended phenotypes: organizational structures that ensure its own replication independently of the beliefs of any particular member. Extended Phenotype Theory (EPT), developed in Dawkins (1982) and applied to institutional analysis in Lerer (2025a, 2025b, 2026), provides a framework in which behavioral states are understood as phenotypic expressions of underlying replicators capable of building structural supports for their own reproduction. Multilevel Evolutionary Game Theory (EGT), particularly the replicator dynamics framework of Nowak (2006), provides formal tools for specifying when a strategy is not merely emergent but evolutionarily stable: resistant to invasion by alternative strategies.

The central claim of this article is threefold. First, CollMD is an evolutionarily stable strategy in organizational environments with high internal normative rigidity, measured through the Organizational Lock-in Index (OLI) introduced here. Second, CollMD achieves stability through the construction of extended phenotypes that reproduce the collective moral state independently of individual turnover. Third, CollMD is the psychosocial mechanism that resolves the intra-corporate intentionality paradox identified in Lerer (2025c): the systematic production of Level 1 organizational behavior by populations of Level 3 individuals. It does so by collectively reconstructing the normative space until Level 3 individuals can operate within Level 1 systems without the cognitive dissonance that would otherwise produce whistleblowing, resignation, or sabotage.

II. THEORETICAL BACKGROUND

2.1 Collective Moral Disengagement: The Schaefer Framework

Epistemological positioning.

This article belongs to a research programme (Lakatos, 1978) that applies Generalized Darwinism and Extended Phenotype Theory to legal, institutional, and organizational systems. Popper (1959) supplies the demarcation criterion: the hypotheses advanced here are falsable and their empirical tests are specified in Section 3.4. Lakatos provides the programme architecture: the hard core (institutions and organizations evolve through variation, selection, and retention mediated by replicators that build extended phenotypes) is protected by an auxiliary belt of specific metrics, including the OLI introduced here, that are individually revisable without abandoning the core commitment. Hull (1988) and Campbell (1974) supply the epistemological grounding for applying evolutionary logic to cultural and normative transmission.

Moral disengagement theory originates with Bandura (1999), who identified eight psychological mechanisms enabling individuals to engage in behavior that violates their own moral standards without experiencing guilt: moral justification, euphemistic labeling, advantageous comparison, displacement of responsibility, diffusion of responsibility, dehumanization, attribution of blame, and disregard for consequences. These mechanisms are individual-level constructs, locating the locus of moral failure in the cognitive processes of the individual actor.

Schaefer et al. (2026) argue that this individual focus is inadequate to explain organizational wrongdoing because the empirical pattern is collective rather than individual. At any of the major corporate scandals documented in the literature, the wrongdoing required coordination, shared justifications, and distributed agency in ways that resist decomposition into individual moral failures summed together. The Schaefer framework introduces CollMD as a group-level construct: a collectively shared set of moral cognitions enabling group members to participate in unethical behavior. CollMD emerges through three phases.

In individual sensemaking, an organizational member encounters a tension between an instruction or practice and prior moral standards. In sensegiving, that member attempts to influence others through argumentation, narrative framing, or appeal to authority, shifting the collectively shared interpretation of what the group is doing and whether it is permissible. In collective sensemaking, the group arrives at a shared reconstruction of both morality ("behavior B is not wrong") and agency ("we are not responsible for behavior B"). The resulting CollMD state is characterized by recursive feedback between individual and collective levels: CollMD shapes subsequent individual moral cognition, which reinforces CollMD, generating a spiral of normalization.

Schaefer et al. draw on complex adaptive systems theory to characterize CollMD dynamics as nonlinear, with tipping points, social cascades, and S-curve diffusion patterns consistent with the Granovetter (1978) threshold model. Three contingencies shape whether CollMD accelerates, stalls, or collapses: the persuasiveness of moral disengagement arguments, the prior proportion of members already morally disengaged, and the relevance of advocates to the collective. This characterization establishes CollMD as a phenomenon with emergent group-

level properties. What it does not provide is a specification of when CollMD is evolutionarily stable, what structural mechanisms sustain it across personnel turnover, or what the transmission mechanism is for new organizational members who join after CollMD has been established.

2.2 Evolutionarily Stable Strategy: Formal Foundations

The concept of evolutionarily stable strategy was introduced by Maynard Smith and Price (1973) and systematized in Maynard Smith (1982) as a solution concept for evolutionary games that does not presuppose the rational deliberation assumed by Nash equilibrium. A strategy s^* is an ESS if and only if two conditions are satisfied: s^* must be a best response to itself, and if an alternative strategy achieves the same payoff against s^* , then s^* must outperform that alternative when the two strategies interact against each other. These conditions jointly ensure that a population playing s^* cannot be successfully invaded by a mutant playing any alternative strategy, provided the mutant population is initially small.

The replicator dynamics framework of Taylor and Jonker (1978) and Nowak (2006) provides the continuous-time dynamical counterpart: strategies with above-average fitness increase in frequency, strategies with below-average fitness decrease. An ESS corresponds to a stable fixed point of the replicator equation. This framework does not require individual actors to consciously choose strategies or compute payoffs; it requires only that the frequency-dependent selection mechanism operates over behavioral variants, however those variants are transmitted. Applied to organizational behavior, ESS analysis asks: given a population of organizational members who can play CollMD, moral compliance, or active dissent, under what conditions is CollMD resistant to invasion by small fractions playing alternative strategies?

2.3 Extended Phenotype Theory Applied to Organizations

Dawkins (1982) extended the standard gene-centered view of evolution by arguing that the phenotypic effects of a gene need not be confined to the body of the organism carrying it. A beaver gene that causes dam-building has as its phenotype not only the beaver body but the dam. These extended phenotypic effects can be just as subject to natural selection as internal phenotypic effects, provided they affect the replication probability of the underlying gene.

Lerer (2025a) applies this framework to legal and institutional analysis by treating normative memes as the relevant replicators. A normative meme achieves phenotypic expression through the institutional structures it generates: courts that enforce it, curricula that transmit it, professional associations that defend it, administrative apparatuses that implement it. These structures constitute the extended phenotype of the meme and are subject to selection because they affect the replication probability of the underlying normative variant. The Constitutional Lock-in Index (CLI) operationalizes this framework by measuring the degree to which a constitutional provision has generated extended phenotypic structures sufficient to resist reform. Empirical validation on 60 international cases yielded AUC 0.97 (Lerer, 2025a). The present article introduces the Organizational Lock-in Index (OLI) as an intraorganizational analogue, measuring the degree to which a CollMD state has generated extended phenotypic

structures within an organization sufficient to resist reform through ethics interventions, compliance programs, or regulatory pressure.

2.4 Heteronomous Bayesian Updating

Standard Bayesian learning models assume agents update beliefs on the basis of evidence regarding the state of the world. Heteronomous Bayesian Updating (HBU), introduced in Lerer (2025b), captures a structurally different process: agents form priors not through direct evidence about the world but through observation of reactions by authority figures to behavioral options. The updating rule is: $P(\text{norm } N \mid \text{authority reacts positively to behavior } B)$ rises, and $P(\text{norm } N \mid \text{authority reacts negatively to behavior } B)$ falls, regardless of the agent's independent assessment of the epistemic or moral content of N .

HBU was originally developed to explain how students in legal education form strong normative priors that predict institutional lock-in without explicit instruction. The mechanism is observational: students observe which arguments generate approval, which positions advance careers, and which challenges to dominant doctrine produce social sanctions. The application to CollMD transmission is direct. New organizational members do not arrive at the moral consensus of their group by being persuaded of its arguments. They arrive there by observing what reactions different behaviors generate. The key empirical implication is that the speed of CollMD transmission to new members is a function of V_r (visibility of reactions to dissent) rather than of argument quality, a prediction that distinguishes the EPT/EGT account from the Schaefer emergentist account.

2.5 Asymmetric Intentionality Theory and the Intra-Corporate Paradox

Asymmetric Intentionality Theory (AIT), developed in Lerer (2025c, 2025d), classifies agents by intentionality level following Dennett (1987, 1989). Level 1 agents are optimizers that pursue fixed objectives through cost-benefit computation without mental state attribution. Level 3 agents engage in full recursive theory of mind, moral reasoning, and identity-based commitment. The Generalized Intentionality Mismatch Theorem predicts systematic failure when regulatory regimes embed Level 3 assumptions about corporate behavior while corporations operate as Level 1 optimizers. Compliance theater, recidivism, and social enforcement failure are the three predicted failure modes, validated across 12 regulatory domains (Lerer, 2025d).

The intra-corporate intentionality paradox, identified in Lerer (2025c), complicates this picture. Corporations contain Level 3 individuals who possess genuine moral reasoning and capacity for shame. The paradox is that these Level 3 individuals, operating within Level 1 organizational systems, produce collective behavior that no individual with Level 3 intentionality would endorse upon full reflection. The filtering architecture converts Level 3 inputs (moral judgments, ethical objections) into Level 1 outputs (cost-benefit calculations) by treating individual judgments as variables in an optimization rather than as decisive constraints. CollMD is the missing psychosocial mechanism that explains how this filtering operates: it reconstructs the moral framework within which individual judgments are formed, so that Level 3 individuals arrive at moral conclusions consistent with Level 1 organizational behavior

without experiencing the dissonance that would accompany direct suppression of their moral reasoning.

III. COLLMD AS EVOLUTIONARILY STABLE STRATEGY: THE FORMAL ARGUMENT

3.1 The Strategic Space

Consider an organization with N members, each of whom can adopt one of three behavioral strategies with respect to a practice P that produces organizational benefits while violating external ethical or legal standards. Strategy C (Compliance): the member treats the external standard as binding, reports concerns through available channels, and declines to participate in P . Strategy M (Moral Dissent): the member participates in P under organizational pressure but actively challenges its acceptability through argumentation, escalation, or external communication. Strategy D (Disengagement): the member participates in P and adopts or generates moral justifications that reconstruct P as acceptable, diffuse responsibility for its consequences, or minimize its harms. These strategies differ in their payoffs along career advancement, cognitive load, and external liability exposure dimensions.

3.2 Payoff Matrix and ESS Conditions

Let $u(x, y)$ denote the fitness payoff to strategy x when interacting with a population predominantly playing strategy y . In a high-OLI organizational environment: $u(D, D)$ is high: the CollMD narrative is validated and reinforced, career advancement proceeds, cognitive load is distributed, and external detection risk is shared. $u(C, D)$ is low to negative: the compliant member is isolated and may face active sanctions for threatening collective narrative coherence. $u(M, D)$ is very low: the dissenting member becomes the focal point for active suppression mechanisms.

Under these conditions, D satisfies the ESS conditions of Maynard Smith (1982): it is the best response to itself, and where alternatives achieve the same payoff against D , D outperforms them when the two interact against each other because the CollMD extended phenotype penalizes C and M more severely in D -dominant environments. D is therefore an ESS in high-OLI organizational environments. The OLI threshold for D to become an ESS is estimated, based on the case analysis in Section VI, at approximately 0.65, analogous to the CLI threshold identified in the constitutional analysis.

3.3 The Organizational Lock-in Index

The OLI formalizes the organizational fitness landscape that determines whether CollMD is an ESS. It is defined as:

$$OLI = w_1 \times P_m + w_2 \times V_r + w_3 \times A_d + w_4 \times D_f$$

where P_m is the proportion of members already operating within the CollMD framework; V_r is the visibility and severity of organizational reactions to moral dissent (normalized 0-1); A_d is the degree of structural isolation experienced by dissenters; and D_f is the density of organizational structures functionally dependent on the maintenance of CollMD, including

performance evaluation systems that decouple results from methods, reporting structures that diffuse accountability, and pre-packaged moral justification narratives. The weights w_1 through w_4 sum to 1 and are subject to empirical calibration. Evidence from the case analysis suggests V_r has the highest predictive value for transmission speed, while D_f has the highest predictive value for persistence across personnel turnover. P_m functions as a threshold variable: below approximately 0.3, CollMD is unlikely to establish as an ESS regardless of other component values.

3.4 Falsifiable Predictions

The EPT/EGT account generates four falsifiable predictions distinguishing it from the Schaefer emergentist account. Prediction 1 (OLI threshold): organizations in which CollMD has stabilized as an ESS will exhibit OLI values above 0.65 when measured prior to the public materialization of compliance failure. Prediction 2 (HBU transmission): the speed with which new organizational members adopt CollMD correlates with V_r rather than with the argumentative quality of available moral justifications. Prediction 3 (extended phenotype persistence): CollMD extended phenotypes are detectable in organizational documents between 18 and 36 months before compliance failures become visible to external parties. Prediction 4 (intervention vector): compliance interventions that target narrative content produce no sustained reduction in CollMD where OLI exceeds 0.65; interventions that alter V_r or D_f produce measurable OLI reduction. Each prediction is falsifiable through empirical adjudication against future organizational data.

IV. THE EXTENDED PHENOTYPE OF COLLECTIVE MORAL DISENGAGEMENT

4.1 From Emergence to Structural Reproduction

The Schaefer framework explains how CollMD emerges. What it does not explain is why CollMD persists after its originators leave the organization. Organizational turnover is substantial: average executive tenure in large corporations is approximately five years. If CollMD were solely a product of the shared cognitive processes of the individuals who generated it, it would dissolve with their departure. Empirically, it does not. The extended phenotype framework provides the explanatory mechanism. The relevant replicator is not any individual's mental state but the normative meme complex encoding the CollMD framework. This meme complex achieves reproductive fitness by constructing organizational structures that ensure its transmission to incoming members and its defense against challenges from external parties.

These structures constitute the extended phenotype of CollMD. They are organizational artifacts, not individual beliefs. A performance evaluation system that rewards revenue generation without auditing method legitimacy persists as an organizational structure long after the executives who designed it have retired. An accountability structure that distributes responsibility across many hands persists as organizational procedure. These structures are the phenotypic expression of the CollMD meme complex, and they are what makes CollMD an ESS rather than a transient emergent state.

4.2 Categories of CollMD Extended Phenotypes

Four categories of CollMD extended phenotype are identifiable in the documented cases. Performance evaluation structures measure outputs without auditing method legitimacy, selecting against members who sacrifice output for ethical constraint and selecting for members who prioritize output and accept available moral justifications. Accountability diffusion architectures are organizational designs that distribute decision-making authority across multiple actors in ways that prevent any individual from being clearly responsible for any outcome. Pre-packaged moral justification narratives are organizational texts and discourses that frame problematic practices as legitimate, necessary, or competitive imperatives, reducing the cognitive cost of D-strategy adoption for new members. Dissent suppression mechanisms include both formal structures (HR procedures that reclassify ethics complaints as performance problems) and informal structures (social exclusion of known dissenters, reputational management that labels ethical objectors as not being team players) that directly increase V_r and A_d components.

4.3 The Volkswagen Case: Extended Phenotype Anatomy

Volkswagen's defeat device program ran from approximately 2006 to 2015, nine years spanning multiple CEO tenures, multiple executive teams, and multiple generations of engineers. The persistence of the program across this turnover is consistent with extended phenotype function rather than individual moral failure. The performance evaluation infrastructure tied advancement to achieving both emissions certification and performance targets that were, according to the forensic record, impossible to achieve simultaneously through legitimate engineering means. The organizational response constructed a technical solution that resolved the impossibility by producing two behavioral modes for the vehicle: one for test conditions, one for road operation.

The moral justification narrative available to engineers and managers reconstructed the defeat device as a legitimate engineering judgment: a calibration choice within accepted parameters, a competitive response to allegedly common industry practices, and a temporary measure pending genuine engineering solutions. The accountability diffusion structure operated through multi-level corporate hierarchy, dispersing knowledge and decision responsibility in ways that made individual prosecutorial attribution extremely difficult. According to DOJ filings and court testimony, engineers who raised concerns were systematically overridden and their concerns were attributed to misunderstanding of management directives. All of these organizational artifacts were detectable in documents well before the September 2015 public disclosure, preceding the materialization of external risk by seven to eight years.

V. HETERONOMOUS BAYESIAN UPDATING AS TRANSMISSION MECHANISM

5.1 The Transmission Problem and HBU Resolution

The emergentist account of CollMD faces a transmission problem: how does CollMD spread to members who join an organization after the initial sensegiving and collective sensemaking

phases have occurred? These members did not participate in the construction of the shared moral narrative. HBU provides a precise account. New members do not primarily update their moral priors by evaluating the arguments for and against CollMD positions. They update by observing what reactions different behaviors produce in the organizational environment. This observational updating mechanism is rapid, reliable, and does not require explicit instruction or persuasion. It requires only that reactions be visible.

Consider a new member joining an organization with established CollMD. In the first weeks and months of organizational membership, the member observes a series of interactions: what happens to colleagues who question certain practices, what happens to colleagues who achieve high performance through methods of ambiguous legitimacy, what topics are raised in performance reviews. Each observation constitutes an HBU update. If raising concern about practice P produces visible negative consequences, the member's prior that "raising concerns about P is normatively appropriate in this organization" receives a strong downward update. This updating mechanism operates independently of whether the member finds the arguments for CollMD persuasive. A member who privately considers a practice morally problematic will nonetheless update toward CollMD compliance if observed reactions to dissent are sufficiently negative, producing behavioral CollMD that contributes to P_m regardless of internal conviction.

5.2 The Central Falsifiable Prediction and Compliance Implications

The HBU account generates a prediction that directly contradicts the sensegiving account in the Schaefer framework. If sensegiving is the primary transmission mechanism, CollMD adoption speed should correlate with argument quality. If HBU is the primary mechanism, CollMD adoption speed should correlate with V_r regardless of argument quality. This prediction is empirically testable through surveys of new organizational members measuring both perceived argument quality and observed reaction severity as predictors of adoption speed.

If HBU is the primary transmission mechanism, the standard architecture of corporate ethics programs is misaligned with the actual causal pathway. Ethics training that presents narrative arguments for ethical behavior, codes of conduct that articulate values, and town hall meetings that communicate leadership commitment to ethics all target narrative content. HBU predicts these interventions have negligible effect when OLI exceeds 0.65, because new members are forming moral priors through observing reactions, not evaluating arguments. The effective intervention vector is V_r : what visibly happens to organizational members who raise ethical concerns. An organization in which raising concerns produces visible career advancement and public recognition will transmit anti-CollMD priors through HBU regardless of the quality of its ethical training content.

VI. CASE ANALYSIS

6.1 Case Selection and Methodology

Four cases were selected on the basis of three criteria: sufficient public documentation of internal organizational dynamics (derived from regulatory proceedings, court filings,

congressional testimony, and forensic analyses); documented temporal trajectory from organizational origin to public materialization of compliance failure; and representation of different industry sectors, regulatory environments, and CollMD content (emissions fraud, bribery, account fraud, and pharmaceutical marketing fraud). The methodology is qualitative case analysis using the EPT/EGT framework as an interpretive template. For each case, the analysis identifies: estimated OLI trajectory based on available documentation; CollMD extended phenotype structures detectable in organizational documents prior to public disclosure; HBU transmission patterns evidenced in accounts of new member adoption; and tipping point dynamics consistent with EGT predictions. The analysis claims that the EPT/EGT framework generates a coherent and empirically consistent interpretation of documented evidence, and that this interpretation generates predictions falsifiable against future case data.

6.2 Volkswagen AG (2006-2015)

Available documentation suggests Volkswagen's OLI crossed the 0.65 threshold approximately in 2008-2009, two to three years after the initial defeat device decision in 2006 and approximately six to seven years before public disclosure in September 2015. P_m was rising by 2007-2008 as the defeat device was deployed across multiple vehicle models and multiple engineers became operationally involved in its maintenance. V_r was high from early in the program: according to DOJ filings and court testimony, engineers who raised concerns were systematically overridden, and in some documented instances their concerns were attributed to misunderstanding of management directives rather than legitimate ethical objection. D_f was extremely high: the entire Clean Diesel commercial strategy, with its associated marketing infrastructure, sales targets, and regulatory certification processes, was structurally dependent on the maintenance of the defeat device program, making dismantlement without full commercial restructuring organizationally impossible. Accounts from engineers indicate that new engineers joining teams working on defeat-device vehicles observed that senior engineers treated the test-road performance discrepancy as a calibration matter, that career advancement proceeded normally for colleagues who managed the device without raising concerns, and that questions about the discrepancy were addressed with technical explanations that did not register it as a compliance problem: a textbook HBU transmission pattern.

6.3 Siemens AG (Late 1990s-2008)

The Siemens bribery system, involving approximately 1.4 billion euros in corrupt payments across multiple divisions and decades, represents one of the most extensively documented cases of institutionalized CollMD in corporate history. OLI indicators suggest stabilization above 0.65 no later than 2002-2003, approximately five to six years before public disclosure in 2006-2008. P_m was very high: the DOJ proceedings indicate involvement by hundreds of managers across multiple divisions and geographic regions. V_r was high in the relevant direction: forensic accounts indicate that managers who refused to participate in payment arrangements were transferred or replaced, while those who participated advanced. D_f was structurally embedded: the bribery system was integrated into project bidding processes, budget structures, and accounting procedures in ways that made extraction without restructuring the entire project

acquisition infrastructure impossible. The Chancellor Schmidt anecdote documented in Schaefer et al. (2026) illustrates the extended phenotype transmission function: when the Chancellor encouraged "exotic thinking" about competitive strategy, the organizational interpretation became encoded in narrative resources subsequently invoked to frame bribery as strategic innovation, entering the D_f component and remaining available as organizational justification long after the originating context was forgotten.

6.4 Wells Fargo (2002-2016)

The Wells Fargo unauthorized account scandal involved approximately 5,300 employees and 3.5 million unauthorized accounts over fourteen years. The OCC enforcement action, CFPB proceedings, and congressional testimony record indicate CollMD was stabilized above 0.65 no later than 2006-2007, approximately nine years before public disclosure. The performance evaluation structure, the "eight is great" sales quota system, created structural pressure impossible to satisfy through legitimate account opening in many markets. V_r was very high and is the most extensively documented component: managers who raised concerns about sales practices were subjected to performance improvement plans, territory changes, and dismissal, as documented in OCC proceedings. Multiple former employees testified to the rapidity with which they understood that the organizational environment rewarded results over method legitimacy, within weeks of joining: a clear HBU transmission pattern. The moral justification narrative reconstructed unauthorized account opening as customer service ("customers would want these products if they understood them"), consistent with organizational values and eventual customer behavior validation.

6.5 Purdue Pharma (1996-2019)

The Purdue Pharma OxyContin case involves a CollMD extended phenotype of particular structural sophistication. The performance evaluation structure rewarded sales achievement through a compensation system in which representatives received bonuses directly tied to OxyContin prescriptions, with no organizational mechanism for auditing whether prescribing patterns were clinically appropriate. D_f was the highest among the four cases: the company commissioned and selectively cited research supporting the low-addiction-risk framing, created educational materials embedding this framing as received scientific consensus, and trained its sales force to present this framing to physicians as authoritative. The CollMD narrative was not merely available as organizational discourse but was embedded in documents presented as scientific evidence. V_r operated primarily at the physician level: physicians who expressed skepticism were identified by sales representatives as targets for additional educational visits, with the organizational objective of resolving skepticism rather than escalating concerns, systematically suppressing external dissent while reinforcing the CollMD framework.

6.6 Cross-Case Comparison

Table 1 summarizes OLI component patterns across the four cases based on available documentation. Values are qualitative assessments, not precise measurements, subject to revision upon availability of more granular organizational data.

Case (year)	P_m	V_r	A_d	D_f	Est. OLI
Volkswagen (2008)	High (0.70)	High (0.80)	Mod-High (0.70)	Very High (0.90)	0.78
Siemens (2002)	Very High (0.85)	High (0.80)	High (0.75)	High (0.80)	0.80
Wells Fargo (2007)	High (0.70)	Very High (0.90)	High (0.80)	High (0.75)	0.79
Purdue Pharma (2002)	Mod-High (0.65)	Moderate (0.65)	High (0.75)	Very High (0.90)	0.74

Table 1. Estimated OLI Components at Approximate Stabilization Date. Values are qualitative assessments based on public documentation. Est. OLI = unweighted mean of four components.

In all four cases the estimated OLI exceeds 0.65 at the identified stabilization date, which precedes public materialization of the compliance failure by between six and twelve years, consistent with Prediction 1. D_f is elevated in all four cases, supporting the prediction that extended phenotype density is the key persistence mechanism. V_r is the dominant component in Wells Fargo; D_f is dominant in Purdue Pharma; both are high in Volkswagen and Siemens. This variation suggests different pathways to CollMD stabilization may exist, with potentially different optimal intervention vectors: V_r-dominant pathways call for reaction architecture reform, while D_f-dominant pathways require structural dismantlement of extended phenotype infrastructure.

VII. IMPLICATIONS

7.1 For Compliance Program Design

The EPT/EGT framework implies that compliance programs should be evaluated not by the sophistication of their narrative content but by their effect on OLI components, particularly V_r and D_f. A compliance program that does not alter what visibly happens to members who raise concerns, and that does not restructure the performance evaluation and accountability architectures constituting the CollMD extended phenotype, will not reduce CollMD in high-OLI environments regardless of its narrative quality. The timing of intervention is critical. Below OLI 0.40, conventional narrative-targeted interventions may suffice to prevent stabilization. Between 0.40 and 0.65, structural interventions targeting V_r and D_f can prevent ESS establishment. Above 0.65, CollMD is established as an ESS, and intervention requires structural disruption: fundamental restructuring of performance evaluation systems, accountability architectures, and leadership composition, equivalent to destroying the extended phenotype rather than updating the narrative.

OLI components are observable in organizational documents, HR data, and governance structures: performance evaluation systems decoupling results from methods; accountability architectures diffusing responsibility; HR outcome data for employees who have filed ethics

concerns; availability of pre-packaged moral justification resources in organizational communications. These indicators can be operationalized as early warning metrics, with OLI monitoring providing advance detection of CollMD stabilization before external compliance failure materializes. In the Argentine regulatory context, Law 27.401 on corporate criminal liability requires compliance programs incorporating codes of ethics, internal reporting channels, training, and due diligence. The EPT/EGT framework suggests these requirements, while necessary, are insufficient in high-OLI organizations because they target narrative content rather than reaction architecture. An amendment incorporating OLI-based risk assessment and mandating structural interventions targeting V_r and D_f would represent a significant regulatory improvement.

7.2 For Corporate Criminal Liability

The CollMD analysis adds a dimension to the AIT account of corporate criminal liability. When CollMD is stabilized above OLI 0.65, the individual Level 3 members of the organization are not suppressing their moral judgments under organizational pressure; they have been incorporated into a collectively reconstructed moral framework through HBU and collective sensemaking. This has a specific implication for individual liability: in high-CollMD environments, the moral reconstruction may be so thorough that the standard assumption underlying personal criminal liability, that the individual knew what they were doing was wrong and chose to do it anyway, may not hold for the majority of participants. The EPT/EGT framework points toward the architects of the extended phenotype, those who designed the performance evaluation systems, accountability architectures, and organizational narratives constituting the CollMD extended phenotype, rather than toward rank-and-file members whose behavior was shaped by HBU transmission of those structures. Regulators and prosecutors can use OLI assessment as an analytical tool: high OLI with concentrated D_f architecture suggests liability concentrated in extended phenotype designers; high OLI with distributed D_f suggests systemic organizational liability; low OLI with high V_r suggests targeted accountability for those controlling the reaction architecture.

7.3 For IntegridAI: An OLI-Based Risk Detection Architecture

The EPT/EGT framework provides the theoretical foundation for an OLI-based risk detection module within the IntegridAI compliance intelligence platform. P_m can be approximated through NLP analysis of organizational communications: the proportion employing collective agency framing ("we decided," "the company's position") versus individual accountability framing in contexts where the latter would be appropriate, detecting shifts toward collective agency consistent with CollMD advancement. V_r can be approximated through HR outcome data: career trajectories of employees who filed internal ethics concerns, the proportion of ethics escalations that resulted in investigation versus reclassification as performance issues. A_d can be approximated through organizational network analysis: the degree to which known concern-raisers are structurally isolated from information flows and decision processes following escalation. D_f can be approximated through structural analysis of governance documents: proportion of performance evaluation criteria that are outcome-based without

method specification, availability of pre-packaged moral justification narratives in organizational communications.

The OLI module would generate a composite risk score updated quarterly or annually, with threshold alerts at 0.40 and 0.65 triggering escalating compliance review requirements. The module would also generate component-level diagnostics identifying which OLI dimension is driving elevated risk, enabling targeted intervention design rather than generic compliance program deployment. The four-case analysis provides provisional calibration anchors: OLI above 0.74 in all four cases at stabilization, D_f dominance in Purdue Pharma, V_r dominance in Wells Fargo, balanced multi-component profile in Volkswagen and Siemens.

VIII. LIMITATIONS AND DIRECTIONS FOR FUTURE RESEARCH

This article faces several limitations qualifying its conclusions. First, the OLI is proposed as a theoretical construct without empirical calibration. The component weights w_1 through w_4 are not determined by the case analysis; the cases illustrate consistency with OLI predictions rather than estimate parameters. Empirical calibration requires longitudinal organizational data with granularity sufficient to measure all four components independently, linked to compliance failure timing outcomes. This is methodologically demanding because organizations with high OLI are also those least likely to make relevant data available to researchers. The most promising methodological avenue is systematic analysis of regulatory proceedings and forensic investigations.

Second, the four cases analyzed share a selection bias toward large, high-visibility compliance failures with extensive public documentation. The OLI threshold of 0.65 is estimated from these high-OLI cases and may not generalize to organizations with more moderate CollMD dynamics. Third, the HBU transmission prediction remains empirically untested. Testing requires survey and interview data from new organizational members measuring both perceived argument quality and observed reaction severity as predictors of CollMD adoption speed. Fourth, the article does not develop a detailed account of CollMD dissolution. The extended phenotype framework predicts that dissolution requires destroying extended phenotype structures rather than changing individual moral beliefs, but the specific mechanisms of dissolution are not analyzed here. Fifth, the article does not address the multilevel character of OLI across organizational subunits, nor the conditions under which high-OLI CollMD spreads from one subunit to others.

Future research should pursue four directions: empirical OLI calibration through regulatory proceedings data; experimental testing of the HBU transmission prediction; agent-based modeling of OLI dynamics and CollMD tipping points building on the Granovetter threshold framework; and development and validation of the IntegridAI OLI monitoring module through longitudinal compliance program outcome studies.

IX. CONCLUSION

Schaefer et al. (2026) establish that collective moral disengagement is an emergent group-level phenomenon with complex systems dynamics, not a simple aggregation of individual moral

failures. This is a necessary reframing of the organizational ethics literature. The present article argues that it is not a sufficient one.

Emergence explains origin; evolutionary stability explains persistence. CollMD persists across personnel turnover, resists ethics training, and reproduces itself in new organizational members who had no role in its construction because it builds extended phenotypes: organizational structures that ensure its own replication independently of the beliefs of any particular member. The Organizational Lock-in Index formalizes the conditions under which CollMD achieves evolutionary stability. Above OLI 0.65, CollMD satisfies the formal ESS conditions of Maynard Smith (1982): it is resistant to invasion by compliance or dissent strategies because the extended phenotype selects against those strategies through differential career consequences. Heteronomous Bayesian Updating provides the transmission mechanism: new members adopt CollMD through observation of reactions, a process that is rapid, reliable, and independent of argument quality.

CollMD is also the missing mechanism in the AIT account of organizational wrongdoing. The intra-corporate intentionality paradox, the systematic production of Level 1 organizational behavior by populations of Level 3 individuals, requires a psychosocial resolution mechanism. CollMD provides it: by collectively reconstructing the moral space until Level 3 individuals can operate within Level 1 systems without unsustainable cognitive dissonance, CollMD enables organizations populated by morally ordinary people to produce collectively extraordinary wrongdoing.

The implications are concrete. Compliance programs targeting narrative content while leaving reaction architecture and extended phenotype structures unchanged will not reduce CollMD in high-OLI organizations. Corporate criminal liability targeting rank-and-file CollMD participants may be systemically misdirected toward those whose moral beliefs were reconstructed through HBU rather than those who designed the extended phenotype. Risk detection systems monitoring for compliance failures after they become visible are detecting the end product of a process observable in organizational structure many years earlier.

Understanding organizational wrongdoing requires understanding the evolutionary ecology of moral strategies. CollMD is not an aberration produced by particularly corrupt individuals. It is an evolutionarily stable strategy in environments where the organizational fitness landscape selects for it. Changing that landscape through structural intervention in reaction architecture, accountability design, and extended phenotype dismantlement is the condition for sustainable compliance. Everything else is narrative.

AI tools were used as research support during the preparation of this paper: locating sources, extracting information from the existing literature, and reviewing drafts. The thesis, analytical framework, arguments, and conclusions constitute the original contribution of the author. This is an independent research project; all intellectual responsibility for the content rests with the author.

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